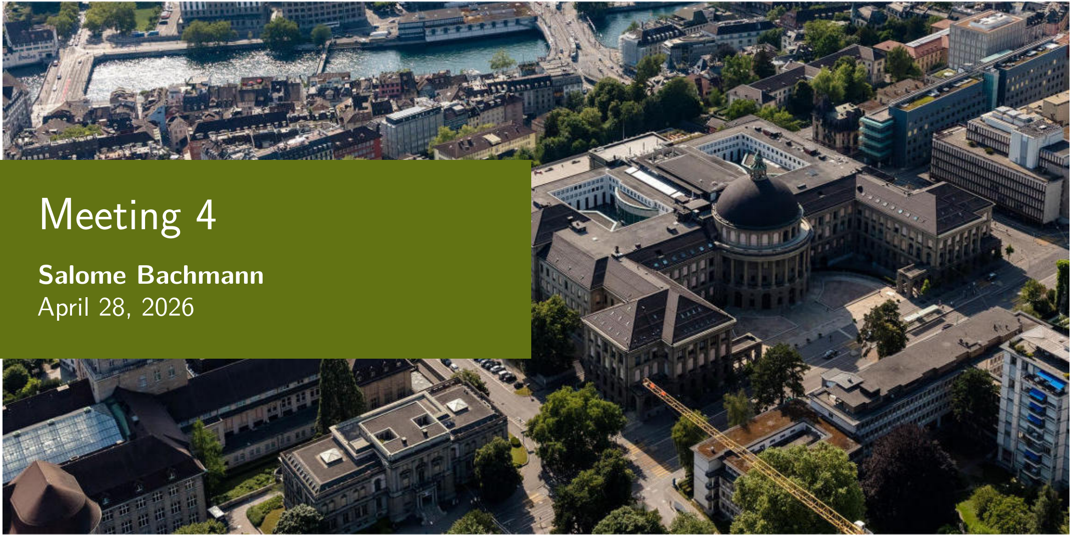

Meeting 4

Salome Bachmann

April 28, 2026

Overview: ToDos of last Meeting

- Compare pcolormesh to imshow in plots
- Plot x and y components of space-time plots
- Source-time function in x and y
- Wiggle plots of traces
- Run supershear simulation to check if it's different from sub-Rayleigh
- Make domain bigger but keep simulation time to get rid of ABC effects
- Summary of crack types

Summary of Model

- Implemented cohesive zone model described by Mahajan and Joshi, 2008
- Implemented functions described by Mahajan and Joshi, 2008
- Accounts for cohesive zone (weak layer)
- Focuses primarily on mode II cracks
- Two-layered model: snow and air
- Fault mechanism: 10Hz moment tensor sources
- Forces of sources are computed by using a fraction of crack modes: eg mxy is calculated by total force times fraction of mode II
- Sources every 2m

Pcolormesh vs imshow

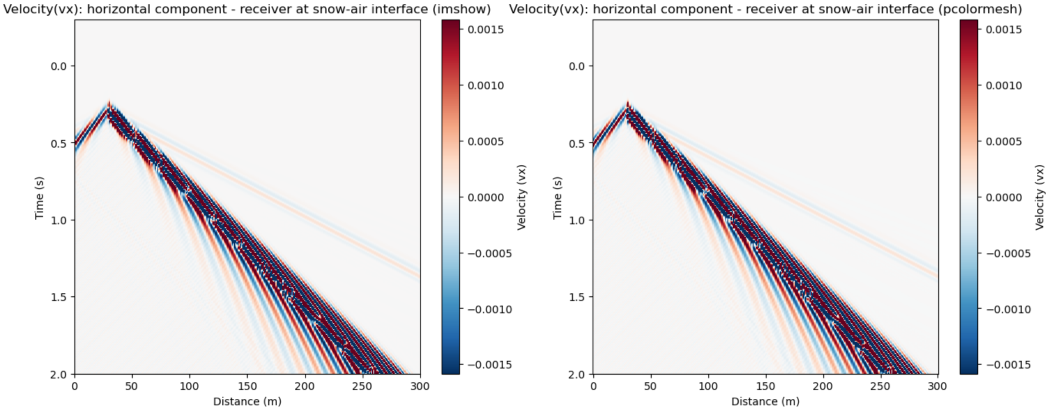

Figure: Comparison of pcolormesh and imshow for plotting space-time plots, this is the supershear case

Sub-Rayleigh vs Supershear

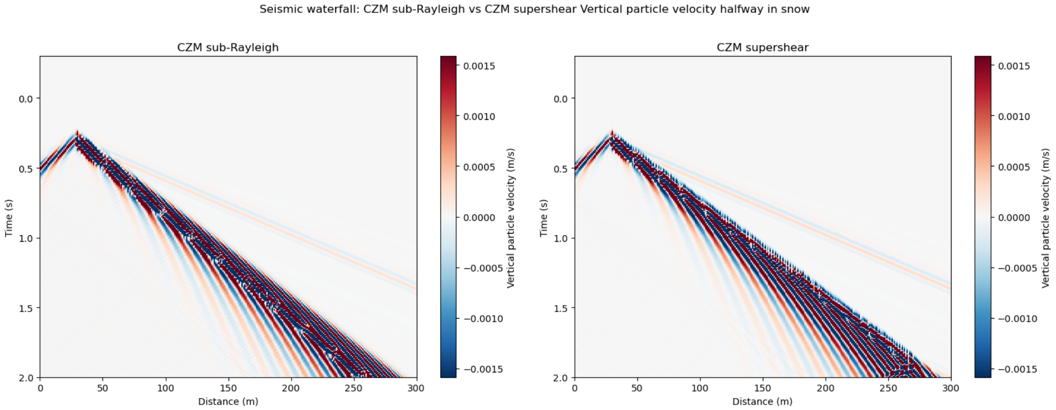

Figure: Comparison of vertical particle velocity sub-Rayleigh and supershear cases

X and y space-time plots

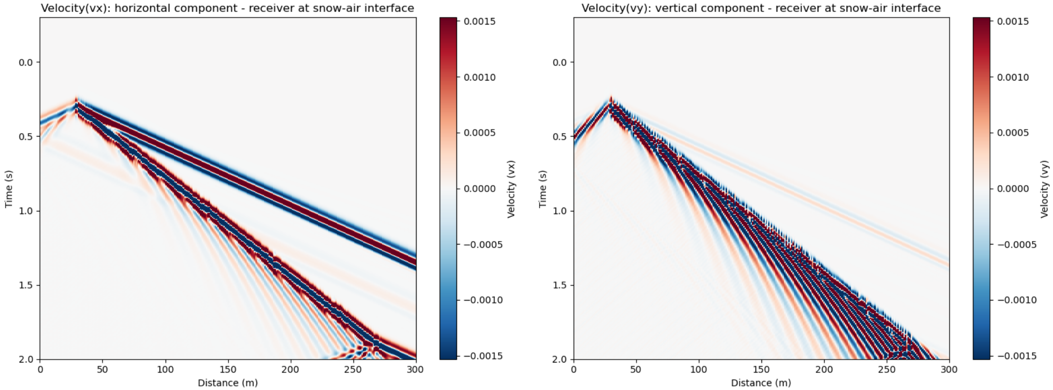

Figure: Space-time plots of vertical and horizontal particle velocity for the supershear case

Bibliography I

Mahajan, P., & Joshi, S. K. (2008). Modeling of interfacial crack velocities in snow. *Cold Regions Science and Technology*, *51*(2), 98–111. https://doi.org/10.1016/j.coldregions.2007.05.008

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